```
                                                                      ___  ____  ____  ____  ____(R)
                                                                     /__    /   ____/   /   ____/
                                                                    ___/   /   /___/   /   /___/
                                                                      Statistics/Data analysis

                                                                        User: printed_result

  ___  ____  ____  ____  ____ ©
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Notes:
      1. Unicode is supported; see help unicode advice.
      2. More than 2 billion observations are allowed; see help obs advice.
      3. Maximum number of variables is set to 5,000; see help set maxvar.

1 . doedit "C:\bivnn_FTU\AK1991_illustration\implementation.do"

2 . do "C:\Users\minhn\AppData\Local\Temp\STD3b18_000000.tmp"

3 . * AK (1991) from Hansen (2022)
4 . use AK1991.dta, clear

5 . * ssc install ivreg2
6 . ivreg2 logwage black smsa married i.yob i.region (edu = i.qob#i.yob), first
Warning - collinearities detected
Vars dropped:       4.qob#1931.yob 4.qob#1932.yob 4.qob#1933.yob 4.qob#1934.yob
                    4.qob#1935.yob 4.qob#1936.yob 4.qob#1937.yob 4.qob#1938.yob
                    4.qob#1939.yob

First-stage regressions
-----------------------


First-stage regression of edu:

Statistics consistent for homoskedasticity only
Number of obs =                 329509
```

edu	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
qob#yob						
1 1931	-.0604458	.052073	-1.16	0.246	-.1625073	.0416157
1 1932	-.2036589	.0501361	-4.06	0.000	-.3019242	-.1053935
1 1933	-.0255154	.0512297	-0.50	0.618	-.1259242	.0748934
1 1934	-.1339962	.0506014	-2.65	0.008	-.2331735	-.0348188
1 1935	-.1431642	.0503901	-2.84	0.004	-.2419274	-.0444009
1 1936	-.160307	.0501249	-3.20	0.001	-.2585503	-.0620636
1 1937	-.1301611	.0494988	-2.63	0.009	-.2271773	-.0331448
1 1938	-.103563	.0483305	-2.14	0.032	-.1982894	-.0088366
1 1939	-.1103965	.048099	-2.30	0.022	-.2046692	-.0161238
2 1930	.1623261	.0491564	3.30	0.001	.0659809	.2586713
2 1931	-.085086	.0520876	-1.63	0.102	-.1871761	.0170042
2 1932	-.1252217	.0509229	-2.46	0.014	-.2250291	-.0254144
2 1933	-.0735227	.0516407	-1.42	0.155	-.174737	.0276915
2 1934	-.0667856	.0509253	-1.31	0.190	-.1665977	.0330264
2 1935	-.0141733	.0503796	-0.28	0.778	-.1129158	.0845693
2 1936	-.1073675	.0505082	-2.13	0.034	-.2063621	-.0083729
2 1937	-.1263954	.0495075	-2.55	0.011	-.2234287	-.0293622
2 1938	-.0395878	.0486513	-0.81	0.416	-.1349429	.0557673
2 1939	-.0990134	.0482647	-2.05	0.040	-.1936109	-.004416
3 1930	.193584	.0486948	3.98	0.000	.0981436	.2890244
3 1931	.0844204	.0515305	1.64	0.101	-.0165779	.1854186
3 1932	-.0932532	.0500496	-1.86	0.062	-.1913489	.0048426
3 1933	.0520994	.0513933	1.01	0.311	-.04863	.1528289
3 1934	-.0829575	.0495376	-1.67	0.094	-.1800498	.0141347

3 1935	.0291449	.0492348	0.59	0.554	-.067354	.1256438
3 1936	-.0399332	.0495641	-0.81	0.420	-.1370774	.057211
3 1937	-.0078533	.0480692	-0.16	0.870	-.1020675	.086361
3 1938	-.0551021	.0474699	-1.16	0.246	-.1481417	.0379376
3 1939	-.1238218	.0468513	-2.64	0.008	-.215649	-.0319946
4 1930	.3369842	.0496198	6.79	0.000	.2397308	.4342375
4 1931	0	(omitted)				
4 1932	0	(omitted)				
4 1933	0	(omitted)				
4 1934	0	(omitted)				
4 1935	0	(omitted)				
4 1936	0	(omitted)				
4 1937	0	(omitted)				
4 1938	0	(omitted)				
4 1939	0	(omitted)				
black	-1.574096	.0205686	-76.53	0.000	-1.61441	-1.533782
smsa	1.048733	.0145985	71.84	0.000	1.02012	1.077346
married	.2252524	.0162537	13.86	0.000	.1933956	.2571092
yob						
1931	.3271538	.0509171	6.43	0.000	.2273577	.4269499
1932	.4725115	.0499273	9.46	0.000	.3746553	.5703676
1933	.4430912	.0503529	8.80	0.000	.344401	.5417814
1934	.5344603	.0496525	10.76	0.000	.4371428	.6317777
1935	.5703313	.0497866	11.46	0.000	.472751	.6679116
1936	.6616281	.049825	13.28	0.000	.5639726	.7592836
1937	.7163029	.04919	14.56	0.000	.6198918	.8127139
1938	.787375	.0487797	16.14	0.000	.6917683	.8829817
1939	.8687366	.048539	17.90	0.000	.7736016	.9638717
region						
1	-1.061263	.0201443	-52.68	0.000	-1.100745	-1.02178
2	-1.67238	.0271911	-61.50	0.000	-1.725674	-1.619087
3	-.6817752	.0210195	-32.44	0.000	-.7229729	-.6405775
4	-.2385156	.0297224	-8.02	0.000	-.2967707	-.1802606
5	-.577982	.0282839	-20.44	0.000	-.6334176	-.5225465
6	-1.097993	.0209826	-52.33	0.000	-1.139118	-1.056868
7	-.7353047	.025661	-28.65	0.000	-.7855995	-.68501
8	-.9498585	.0239234	-39.70	0.000	-.9967476	-.9029694
_cons	12.152	.0424115	286.53	0.000	12.06887	12.23512

F test of excluded instruments:
  F(30,329458) = 4.75
  Prob > F = 0.0000
Sanderson-Windmeijer multivariate F test of excluded instruments:
  F(30,329458) = 4.75
  Prob > F = 0.0000

Summary results for first-stage regressions

Variable	F(30,329458)	P-val	(Underid) SW Chi-sq(30)	P-val	(Weak id) SW F(30,329458)
edu	4.75	0.0000	142.44	0.0000	4.75

Stock-Yogo weak ID F test critical values for single endogenous regressor:

5% maximal IV relative bias	21.42
10% maximal IV relative bias	11.32
20% maximal IV relative bias	6.09
30% maximal IV relative bias	4.29
10% maximal IV size	86.17
15% maximal IV size	44.78
20% maximal IV size	30.72
25% maximal IV size	23.65

Source: Stock-Yogo (2005). Reproduced by permission.
NB: Critical values are for Sanderson-Windmeijer F statistic.

Underidentification test
Ho: matrix of reduced form coefficients has rank=K1-1 (underidentified)
Ha: matrix has rank=K1 (identified)
Anderson canon. corr. LM statistic Chi-sq(**30**)=**142.38** P-val=**0.0000**

Weak identification test
Ho: equation is weakly identified
Cragg-Donald Wald F statistic **4.75**

Stock-Yogo weak ID test critical values for K1=1 and L1=30:

5% maximal IV relative bias	**21.42**
10% maximal IV relative bias	**11.32**
20% maximal IV relative bias	**6.09**
30% maximal IV relative bias	**4.29**
10% maximal IV size	**86.17**
15% maximal IV size	**44.78**
20% maximal IV size	**30.72**
25% maximal IV size	**23.65**

Source: Stock-Yogo (2005). Reproduced by permission.

Weak-instrument-robust inference
Tests of joint significance of endogenous regressors B1 in main equation
Ho: B1=0 and orthogonality conditions are valid

Anderson-Rubin Wald test	F(**30,329458**)=	**1.42**	P-val=**0.0643**
Anderson-Rubin Wald test	Chi-sq(**30**)=	**42.54**	P-val=**0.0642**
Stock-Wright LM S statistic	Chi-sq(**30**)=	**42.54**	P-val=**0.0643**

Number of observations	N =	**329509**
Number of regressors	K =	**22**
Number of endogenous regressors	K1 =	**1**
Number of instruments	L =	**51**
Number of excluded instruments	L1 =	**30**

IV (2SLS) estimation

Estimates efficient for homoskedasticity only
Statistics consistent for homoskedasticity only

			Number of obs =	**329509**
			F(21,329487) =	**1436.03**
			Prob > F =	**0.0000**
Total (centered) SS =	**151837.8708**		Centered R2 =	**0.1584**
Total (uncentered) SS =	**11621827.82**		Uncentered R2 =	**0.9890**
Residual SS =	**127781.0113**		Root MSE =	**.6227**

logwage	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
edu	.0805518	.0163846	4.92	0.000	.0484385	.112665
black	-.2302179	.0261251	-8.81	0.000	-.2814221	-.1790136
smsa	.1581466	.0174229	9.08	0.000	.1239985	.1922948
married	.2439689	.0048706	50.09	0.000	.2344227	.2535152
yob						
1931	-.0064838	.0054362	-1.19	0.233	-.0171385	.0041709
1932	-.0118228	.0058059	-2.04	0.042	-.0232022	-.0004435
1933	-.0128383	.0065003	-1.98	0.048	-.0255787	-.000098
1934	-.0162055	.0068249	-2.37	0.018	-.029582	-.002829
1935	-.0271844	.0077269	-3.52	0.000	-.0423289	-.0120399
1936	-.0275322	.0083179	-3.31	0.001	-.0438349	-.0112294
1937	-.0313496	.0092155	-3.40	0.001	-.0494117	-.0132875
1938	-.0319571	.0104142	-3.07	0.002	-.0523685	-.0115457
1939	-.0412408	.0110988	-3.72	0.000	-.0629941	-.0194875
region						
1	.0343439	.0178348	1.93	0.054	-.0006118	.0692995
2	-.1355007	.0279182	-4.85	0.000	-.1902194	-.0807821
3	-.0409571	.011911	-3.44	0.001	-.0643023	-.017612
4	-.0879569	.0070079	-12.55	0.000	-.1016921	-.0742216
5	-.1033678	.0109775	-9.42	0.000	-.1248833	-.0818523
6	-.1203118	.0184479	-6.52	0.000	-.1564689	-.0841546

```
           7 |  -.0950523   .0130463    -7.29   0.000    -.1206226   -.0694819
           8 |  -.0867657   .0162276    -5.35   0.000    -.1185711   -.0549602
             |
       _cons |    4.62685   .2020062    22.90   0.000     4.230925    5.022775
------------------------------------------------------------------------------
Underidentification test (Anderson canon. corr. LM statistic):         142.381
                                                   Chi-sq(30) P-val =    0.0000
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                4.747
Stock-Yogo weak ID test critical values:  5% maximal IV relative bias    21.42
                                         10% maximal IV relative bias    11.32
                                         20% maximal IV relative bias     6.09
                                         30% maximal IV relative bias     4.29
                                         10% maximal IV size             86.17
                                         15% maximal IV size             44.78
                                         20% maximal IV size             30.72
                                         25% maximal IV size             23.65
Source: Stock-Yogo (2005).  Reproduced by permission.
------------------------------------------------------------------------------
Sargan statistic (overidentification test of all instruments):          22.487
                                                   Chi-sq(29) P-val =    0.7995
------------------------------------------------------------------------------
Instrumented:         edu
Included instruments: black smsa married 1931.yob 1932.yob 1933.yob 1934.yob
                      1935.yob 1936.yob 1937.yob 1938.yob 1939.yob 1.region
                      2.region 3.region 4.region 5.region 6.region 7.region
                      8.region
Excluded instruments: 1b.qob#1931.yob 1b.qob#1932.yob 1b.qob#1933.yob
                      1b.qob#1934.yob 1b.qob#1935.yob 1b.qob#1936.yob
                      1b.qob#1937.yob 1b.qob#1938.yob 1b.qob#1939.yob
                      2.qob#1930b.yob 2.qob#1931.yob 2.qob#1932.yob
                      2.qob#1933.yob 2.qob#1934.yob 2.qob#1935.yob
                      2.qob#1936.yob 2.qob#1937.yob 2.qob#1938.yob
                      2.qob#1939.yob 3.qob#1930b.yob 3.qob#1931.yob
                      3.qob#1932.yob 3.qob#1933.yob 3.qob#1934.yob
                      3.qob#1935.yob 3.qob#1936.yob 3.qob#1937.yob
                      3.qob#1938.yob 3.qob#1939.yob 4.qob#1930b.yob
Dropped collinear:    4.qob#1931.yob 4.qob#1932.yob 4.qob#1933.yob
                      4.qob#1934.yob 4.qob#1935.yob 4.qob#1936.yob
                      4.qob#1937.yob 4.qob#1938.yob 4.qob#1939.yob
------------------------------------------------------------------------------

7 .
  end of do-file

8 .
```